

Tables and Figures

Pipe Selection and Hydraulic Profile

Pipe Specifications	
Pipe Material	HDPE 100
Pipe Length (km)	47
Pipe Size (mm)	630
Pipe Pressure Rating (bar)	PN16
Manufacturer	Welbrit., Ltd [1]

Table 1: Summary of pipeline design specifications and material selection.

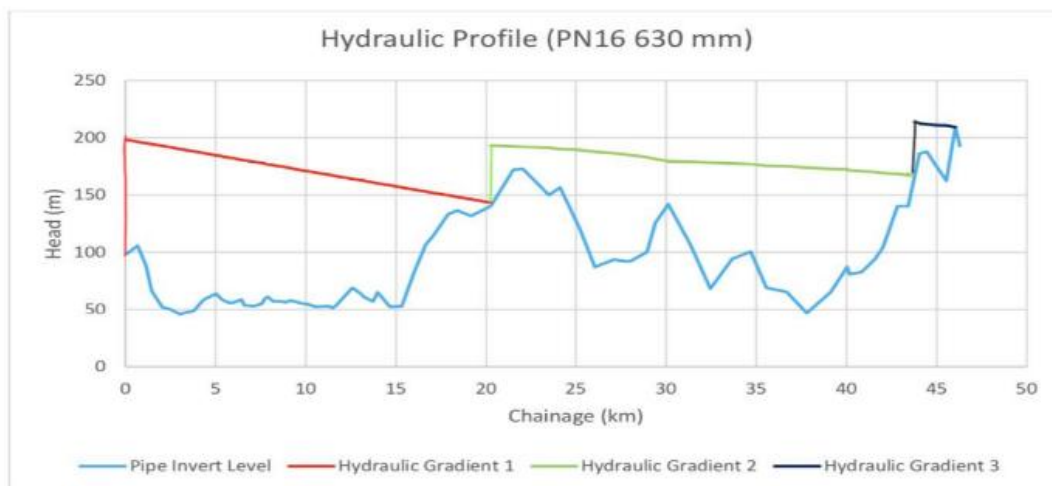


Figure 1: Hydraulic profile for the treated HDPE 100 pipeline (630mm DN / 16 bar PN).

Selected Pump Profile

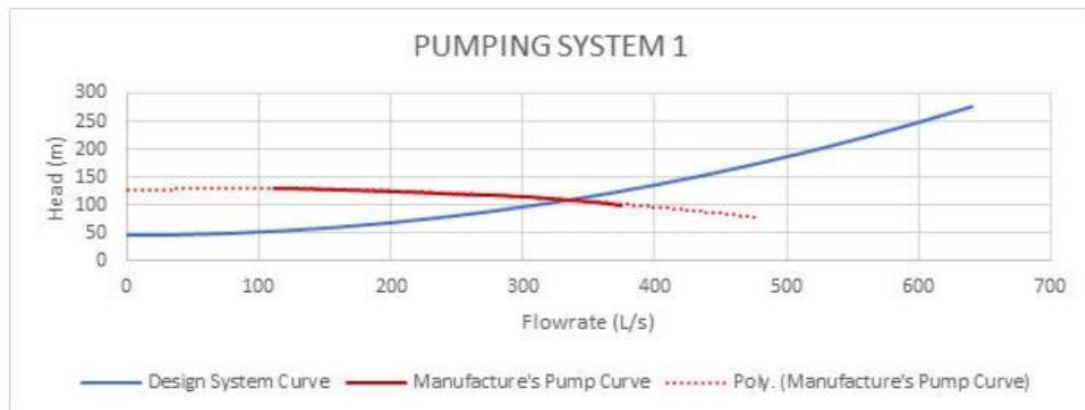


Figure 2: Selected pump for pump station 1 located at the water treatment plant.

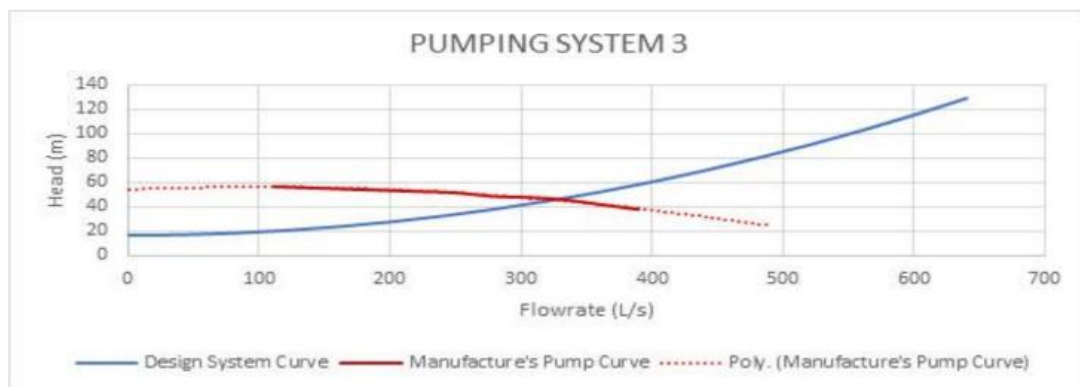


Figure 3: Selected pump for pump station 2 located at the midway of the treated water pipeline route.

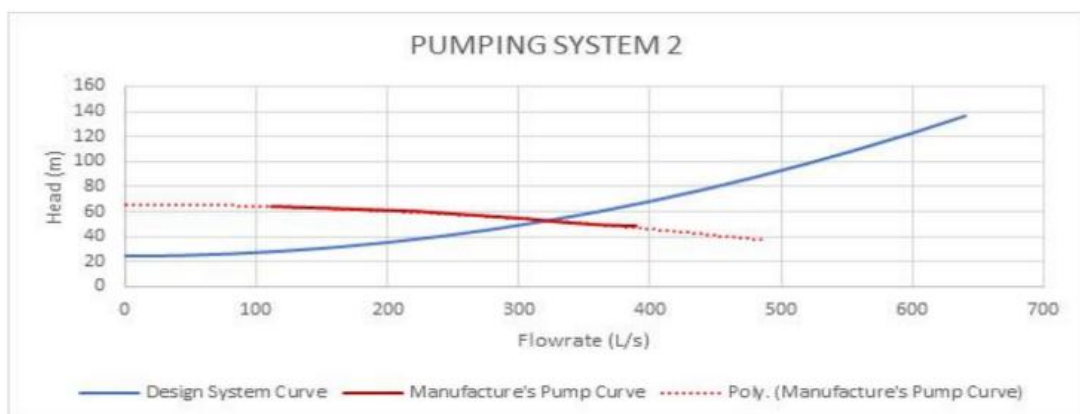


Figure 4: Selected pump for pump station 3 located not far from the water storage tank.

Pump Station Layout and Implementation

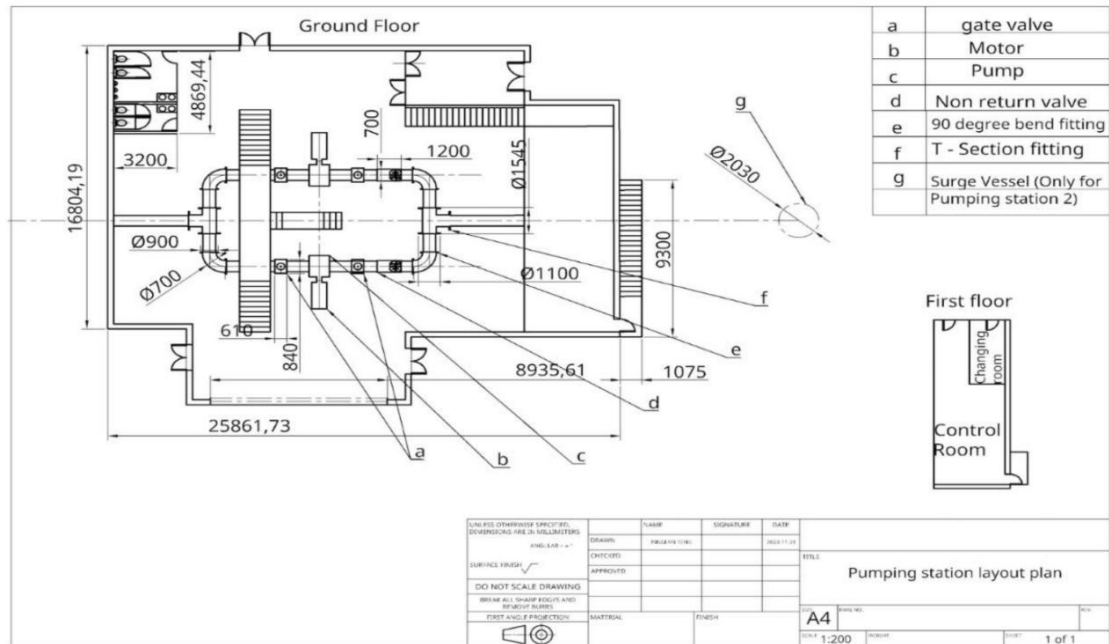


Figure 5: A detailed layout of the 3 pumping stations (One pump serves as a back-up pump and will be automatically switched on when the main pump stops operating).

	Static Head (m)	Duty Flowrate (l/s)	Total Head (m)	Power rating (kw/h)	Pump& Motor Overall Dimension (Width*Length mm)
Pump 1	44.88	312.5	100	444.6	1030 x 3165
Pump 2	24.21	312.5	50	201.8	1160 x 3084
Pump 3	16.59	312.5	45	183.9	1030 x 2795
Pipe Fittings	T Section	90-Degree Bending	Gate Valves	Non-Return Valves	Surge Vessel
Manufacture	JD pipes [2]	Ftductile.,ltd [3]	Avk[4]	Avk[5]	Abbott Pressure Vessels [6]

Table 2: Summary of pump operating conditions, hydraulic head requirements, equipment dimensions, and selected pipe fittings for the proposed water supply system.

General Information of Pump stations	
Installment Instructions	Based on DfMA analysis to solve the efficiency of assembling, Pumps are suggested to be pre-installed and then transported by lorries to the sites (Following the UK max load limit regulations [7]). Cranes are used to displace the pumps on site. Valves and Bend sections are installed on site before the constructing the station housing.
Maintenance	- Detailed pump operation and maintenance instruction is supplied by the manufacture [8] - A 17000 Liter surge vessel is installed along the pipeline to maintain pressure below 16 bar along the pipeline and prevent pipe bursting [6] - Several air valves are used to release accumulated air when the pipe instantly starts filling or draining during pump operation.
Safety& Hazards	- 3 fire escape doors are located on the ground floor and 1 fire escape door on the first floor to access the outdoor stairs - Necessary fire extinguishers and fire blankets are installed in each pumping station. - Outer walls of the pump stations are Wall Type B cavity wall with cavity fill insulation [9]

Table 3: Design, installation, maintenance, and safety provisions for pump stations.

Carbon Analysis

DETAILED CARBON EVALUATION			
Capital Carbon Emission			
Activity	Weight/Volume	Emission Factor [10]	Total Carbon (tCO ₂ e/t)
HDPE 100 Pipe manufacturing	4732.65 tonnes	2.52 tCO ₂ e/t: HDPE material carbon factor	=4732.65 x 2.52 = 11926.3
Transportation of the pipes	4732.65 tonnes (Round trip of 345 km)	0.107 tCO ₂ e/ /tonnekm: all averaged, HGVs	= 4732.65 x 0.107 x 345 / 1000 = 174.71
Installing the pipeline		Assume 3% of the material cost	= 0.03 x 11926.3 = 357.8
Pump & Motor	= 3553.3 x 2 + 4221.6 x 2 + 4596.6 x 2 = 24.743 tonnes	2.80 tCO ₂ e/t: Steel manufactured, averaged	= 24.743 x 2.80 = 69.3
All the Pipe fittings, valves and vessels		Assume 35% of the pump manufacturing	= 0.35 x 69.3 = 24.255
Transportation of pumps & motor, pipe fittings and valves		About 2% of the transportation of pipes	= 0.02 x 174.71 = 3.5
Total			12556
Operational Carbon Emission			
Activity	Power	Emission Factor	Total Carbon (tCO ₂ e/t)
Pumping operation	Average of 3656641.2 kWh/yr	0.091 kgCO ₂ e/kWh: UK commercial grid average 2026	=3656.6 x 0.091 = 332.75
Maintenance (Employee's travel, station lights, annual mechanical& electrical services)		Assume 5% of Pumping operation	= 0.05 x 332.75 = 16.64
Total			350

Table 4: life cycle carbon emissions assessment of pipeline infrastructure showing capital and operational carbon breakdown.

Reference

- [1] - Popalyar, A. (no date) *Polyethylene Pipe, Welbrit Ltd.* Available at: <http://www.welbrit.co.uk/polyethylene.php> (Accessed: 1 November 2023).
- [2] - Ltd, J.D. (Pipes) (no date) *Home, JDP.* Available at: <https://www.jdpipes.co.uk/products-and-services/polyethylene-pressure-systems> (Accessed: 3 November 2023).
- [3] - (No date b) *Technical specifications for ductile iron pipes and fittings.* Available at: https://ftductile.co.uk/wp-content/uploads/2019/09/FTD-book-revised-aug17_FTComp.pdf (Accessed: 3 November 2023).
- [4] - AVK resilient seated gate valve, PN16 (no date) * - AVK UK. Available at: <https://www.avkuk.co.uk/en-gb/product-finder/gate-valves/resilient-seated-gate-valves/21-50-002> (Accessed: 3 November 2023).
- [5] - *Non-return valves guidance paper on the importance of correct selection* (no date) *Non-return valves technical introduction* - AVK UK. Available at: <https://www.avkuk.co.uk/en-gb/case-studies/check-valve-technical-introduction> (Accessed: 3 November 2023).
- [6] - Heather (2012) *Air receivers and pressure vessels manufactured by Abbott and Co (Newark) Ltd, Air Receivers.* Available at: <https://www.air-receivers.co.uk/air-receivers/> (Accessed: 5 November 2023).
- [7] - *HGV maximum weights* (no date) GOV.UK. Available at: <https://www.gov.uk/government/publications/hgv-maximum-weights/hgv-maximum-weights> (Accessed: 10 November 2023).
- [8] - (No date a) *E-XC single stage, double suction centrifugal pumps (50 Hz) - xylem.* Available at: https://www.xylem.com/siteassets/brand/xylem/e-xc-single-stage/brexc_12pg_50hz.pdf (Accessed: 12 November 2023).
- [9] - The Scottish Government (2019) *Building standards technical handbook 2019: Domestic, Scottish Government.* Available at: <https://www.gov.scot/publications/building-standards-technical-handbook-2019-domestic/> (Accessed: 15 November 2023).
- [10] - Department for Energy Security and Net Zero (2023) *Green book supplementary guidance: Valuation of energy use and greenhouse gas emissions for appraisal, GOV.UK.* Available at: <https://www.gov.uk/government/publications/valuation-of-energy-use-and-greenhouse-gas-emissions-for-appraisal> (Accessed: 18 November 2023).